

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 1: Short Answer Test**

**COURSE NAME:**

Cloud Computing and  
Big Data Analytics

**COURSE CODE:**

CSA1516

**COURSE OUTCOME COVERED**

**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)  
Question Count: 5

**Duration: 60 Minutes**

**Total Marks: 25**

**Code of Conduct**

I A. Satwika (192525204) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Satwika

| Criteria  | Conceptual Accuracy (1.5) | Coverage of Concepts (1) | Use of Examples (0.75) | Comparison / Differentiation (1) | Clarity of Expression (0.75) | TOTAL (5) |
|-----------|---------------------------|--------------------------|------------------------|----------------------------------|------------------------------|-----------|
| Weightage | 30%                       | 20%                      | 15%                    | 20%                              | 15%                          | 100%      |
| Q1        |                           |                          |                        |                                  |                              |           |
| Q2        |                           |                          |                        |                                  |                              |           |
| Q3        |                           |                          |                        |                                  |                              |           |
| Q4        |                           |                          |                        |                                  |                              |           |
| Q5        |                           |                          |                        |                                  |                              |           |

**Signature of the Faculty**

**Total Marks Awarded:**

# ASSESSMENT - 01

CSA1516 - Cloud computing and  
Big Data Analytics.

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DATE : 20.07.26

# Evolution from standalone Computing to

## Cloud - Based Systems :-

### Introduction

Computing has evolved from standalone computers to cloud-based systems to meet the increasing demand for storage, processing power, and accessibility. Cloud computing allow users to access computing resources over the internet instead of relying only on local devices.

### Evolution of computing :-

#### a) standalone computing

- \* A single computer performs all processing and stores data locally.
- \* NO resource sharing
- \* Limited storage and computing power.

## b) Client - Server computing

- \* Clients request services from centralized users.
- \* Resources such as databases and appliances are shared.
- \* Easier management than standalone systems.

## c) Distributed computing.

- \* Multiple computers work together to complete tasks.
- \* Improves performance and reliability
- \* Used in scientific and enterprise applications.

## d) cloud computing.

- \* Computing resources are delivered over internet.
- \* Users pay only for resources they use
- \* Resources can be accessed from anywhere.

Example:-

A student storing assignments in Google drive instead of saving them in personal laptop.

# Working of Hypervisors in Virtualisation and Difference between Type 1 and Type 2 Hypervisors:-

## Introduction:-

Virtualisation allows multiple operating systems to run on a single physical computer using a software layer called a hypervisor.

## Working of a Hypervisor:-

- \* Creates and manages VM's.
- \* Allocates CPU, memory, and storage.
- \* Keep each VM isolated and secure.

## Difference b/w Type 1 and Type 2:-

### TYPE 1

- \* Runs directly on hardware
- \* High performance
- \* More Secure
- \* Used in datacentres

Ex:- Microsoft hyper-V,  
Xen

### TYPE 2

- \* Runs on an operating system
- \* Moderate performance
- \* Less Secure
- \* Used in PC & testing

Ex:- Oracle Virtual Box,  
VMware workstation



Application programming interface (API) is a set of rules that enables different software applications to communicate with each other.

Use of API :-

- connect different cloud services
- enable data exchange
- Automate cloud operations
- \* support third-party app integration.
- Simplify app development.

examples:-

\* food delivery app uses google maps API for navigation and payment APIs for online transactions

Advantages:-

- \* faster integration
- \* Better scalability
- \* improved automation
- \* easy maintenance.

# Resource pooling in cloud computing :-

## Introduction :-

Resource pooling is a cloud computing feature where computing resources are shared among multiple users while keeping their data secure and isolated.

## Working :-

- \* Resources such as CPU, memory, storage and networking are pooled together.
- \* Resources are allocated based on user demand.
- \* Users are unaware of exact physical location of resources.

## Advantages :-

- \* High availability
- \* Reduced operational cost
- \* Better scalability
- \* Improved performance
- \* supports multiple users simultaneously.

\*

# Rapid elasticity in cloud computing:-

## Introduction:-

Rapid elasticity is the ability of cloud computing to automatically increase or decrease computing resources according to workload demand.

## Working:-

- \* Resources scale up when demand increases.
- \* Resources scale down when demand decreases.
- \* scaling is automatic and fast

## Advantages:-

- \* Handles sudden traffic increases
- \* Reduces operational costs.
- \* ensures high availability
- \* prevents resource wastage

Example :- During sales like Amazon Great Indian festival, cloud servers gets increased automatically to handle millions of users.

## Applications:-

- \* E-commerce
- \* Online gaming
- \* social media platforms.